CLASS 8 — MATHS · FORMULA SHEET

Chapter 6: **We Distribute, Yet Things Multiply** | Quick revision, one page

Distributive property

- $a(b+c) = ab + ac$ and $a(b-c) = ab - ac$.
- $(a+b)(c+d) = ac + ad + bc + bd$ (each term of one bracket times each term of the other).
- $(x+a)(x+b) = x^2 + (a+b)x + ab$.

Standard identities

- $(a+b)^2 = a^2 + 2ab + b^2$.
- $(a-b)^2 = a^2 - 2ab + b^2$.
- $(a+b)(a-b) = a^2 - b^2$.
- $(a+b)^2 - (a-b)^2 = 4ab$ and $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$.

Factorisation

- Take out the common factor: $ab + ac = a(b+c)$.
- $a^2 - b^2 = (a+b)(a-b)$.
- $a^2 + 2ab + b^2 = (a+b)^2$ and $a^2 - 2ab + b^2 = (a-b)^2$.

Quick calculation

- $99^2 = (100-1)^2$, $102^2 = (100+2)^2$.
- $a \times b = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$, e.g. $53 \times 47 = 50^2 - 3^2 = 2491$.
- Like terms: same variable part (e.g. $3x^2$ and $5x^2$); only like terms can be added or subtracted.

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